MANVIK TALWAR

AI/ML Engineer · Backend Systems · Production AI

manvik.talwar@gmail.com · +91 8800287735 · linkedin.com/in/manvik-talwar · github.com/CodexManvik · codexmanvik.github.io

TECHNICAL SKILLS

AI / ML: LLMs, RAG Pipelines, LangGraph, Hugging Face Transformers, PyTorch, TensorFlow, OpenCV, Scikit-learn, LLM Quantization
Backend: FastAPI, Flask, Python, Node.js, PostgreSQL, MySQL, MongoDB, SQLite, ChromaDB, FAISS, Redis
Cloud & DevOps: Azure OpenAI Service, Azure Cognitive Search, Azure App Service, Docker, CI/CD, Git, Linux
Frontend: React, Next.js, TypeScript, JavaScript, Streamlit, Tailwind CSS

EXPERIENCE

Backend Engineering Intern | *TechLearn* *Mar 2026 – Present*

- Developed Dashboard Stats API aggregating real-time KPIs across student/educator data — adopted as primary analytics endpoint
- Built Filtered Submission API with cursor-based pagination, reducing submission list load time by 5×
- Implemented trend, completion, and volume analytics APIs with query-level optimizations yielding 40% faster response times
- Formulated Submission schema with composite indexes, improving throughput and reliability under high-concurrency conditions

AI Development Intern | *Path Infotech* *Jun 2025 – Aug 2025*

- Architected and deployed 'Sofia' — an enterprise RAG chatbot using Azure OpenAI, Azure Cognitive Search, and Flask backend; live on company servers, serving 100+ concurrent users at 99.9% uptime
- Engineered semantic search pipeline with custom skillsets for PDF text/image indexing using Azure OpenAI embeddings, reducing API response latency by 25% via optimized routing and payload compaction
- Built citation management system with MySQL tagging, enabling clickable source references in every LLM response
- Automated deployment via Azure App Service CI/CD, eliminating manual releases and cutting release lead time

PROJECTS

Agentic RAG for Distributed Content | *Personal Project* *Github Link*

- Built a multi-stage agentic RAG pipeline (normalize → plan → retrieve → adequacy check → synthesize → validate → finalize) using LangGraph, FastAPI, ChromaDB, and Ollama llama3.2:3b running fully locally
- Created 6 specialized agents for domain routing, citation validation, and answer adequacy; achieved <30s end-to-end query latency on constrained hardware (4–8GB VRAM)
- Replaced naive chunking with PyMuPDF ingestion + bibliography filtering, significantly improving retrieval precision; evaluated against 40-question benchmark with Hit@k tracking

Interview Mirror | *Academic Project* *Github Link*

- Constructed a real-time AI interview coach tracking 543 body landmarks (face/hand/pose) using MediaPipe at <100ms latency, with posture and stress scoring overlaid on live video
- Integrated Google Gemini for conversational question generation, faster-whisper for STT, and pyttsx3 for TTS; persisted session data in SQLite with JWT-auth FastAPI backend
- Delivered full WebSocket-based frontend-backend communication for live feedback streaming without polling

Deep Learning Watermarking (DWT + CNN) | *Research Project* *Github Link*

- Designed a CNN encoder-decoder to embed invisible binary watermarks in the frequency domain using Discrete Wavelet Transform, achieving PSNR > 40dB with 96% extraction robustness
- Validated resilience against JPEG compression, Gaussian noise, and cropping attacks using OpenCV-based evaluation pipeline

EDUCATION & ACHIEVEMENTS

B.Tech in Computer Science — AI/ML | *Manipal University Jaipur* *Aug 2023 – Aug 2027*

- 4× Student Excellence Award for academic performance and project execution
- Selected for SIH 2025 All India Round — national-level software innovation competition
- Finalist: Dell × Google Developer Group Ideathon (Team Moriarty)
- Top 15,000 — Google The Big Code Challenge

CERTIFICATIONS

Azure AI Certification (Microsoft) · Azure AI Fundamentals: Generative AI, NLP, CV, Document Intelligence (Microsoft) · Machine Learning Practitioner (Coursera) · Backend Development (Coursera) · Design & Analysis of Algorithms (NPTEL) · Red Hat System Administration I & II (RH124/RH134)